

Application of Random Matrix Theory in Estimating Realistic Implied Correlation Matrix from Option Prices

T.C Urama*, C.P Nnanwa, P.O. Ezepue

Statistics and Information Modelling Research Group, Department of Engineering & Mathematics,
Sheffield Hallam University, S1 1WB, United Kingdom

Abstract

We analyze the cross-correlation matrix C of the stock index returns in the Nigerian Stock Market (NSM) from the period 2009 to 2013 using the concept of random matrix theory. We test the eigenvalues of the selected assets from the exchange and use their respective eigenvector to determine the stocks that drive the market through the analysis of their inverse participation ratios. We also proffer a method of obtaining a realistic implied correlation matrix for a hypothetical portfolio of some given assets selected from those considered in the empirical correlation matrix of the assets considered. We discover that the implied correlations matrix is positive meaning that the corresponding assets move in the same direction. Hence, portfolio diversification is not an optimal investment strategy, rather investment on derivative assets like call and put options is recommended.

Key words:

Random matrix, cross-correlations, implied volatility, portfolio weights, a basket of options.

1.0 Introduction

In this work, we investigate the spectral properties of the correlation matrix of the price fluctuations in an emerging market, Nigerian Stock Market (NSM), by studying the dynamics of stock price movements and trends in the fluctuations in the NSM using the Random Matrix Theory (RMT). We analyze the correlation matrix using Random Matrix Theory (RMT), through a comparison of the empirical correlation matrix with that of the Wishart random matrix. The linear relationships among assets are always summarized in a correlation matrix hence the need to study RMT in any given financial market.

Szilard Pafka and Imre Kondor [1] assert that correlation matrices of financial returns play a crucial role in various aspects of modern finance including investment theory, capital allocation, and risk management. In their opinion, for a theoretical perspective, the main interest in studying correlation of price returns is for proper description of the structure and dynamics of correlations whereas for a practitioner, the emphasis is on the ability of the models to provide adequate inputs towards the numerous portfolios and risk management procedures required in the financial industry. Kawee Numpacharoen [2] notes that financial

institutions usually hold multiple assets and asset classes in their portfolios that may include basket of derivatives, credit derivatives or other correlation trading products which depends largely on the correlation coefficients between the underlying assets.

In this regard, RMT enables us propose suitable derivative pricing models to be applied in the NSM, for portfolio optimization, including risk management, and appropriate pricing formulae for the proposed pioneer derivative products due for introduction in the NSM. A. Sensoy et al. [3] assert that high correlation implies that the benefits of portfolio diversification is lowered since according to their finding, high correlation is synonymous to high volatility of stock prices. In this respect, the better option available to investors is thinking through the derivative (option) trade in risk management of their portfolio of investments. Thus, the need for a comprehensive analysis of the nature of correlation among stocks in a given exchange and relating the observed relations to derivative trade is of utmost importance to investors in any given stock exchange, whether developed and or emerging market(s). It is noteworthy that following the introduction of RMT into the financial markets by R.N. Mantegna [4], Laloux et al. [5] and Plerou et al. [6], it has been used in the study of the statistical properties of cross-correlation in different financial markets [7-32]. Laurent Laloux et al. [5] note that for financial assets, the study of the empirical correlation matrix is very relevant since, from their finding, the estimation of the correlations between the price movements of different assets constitutes an important and indispensable aspect of risk management. They declare that the probability of large losses for a certain portfolio or option book is dominated by correlated moves of its different components and that a position which is simultaneously long in stocks and short in bonds will be risky as stocks and bonds usually move in opposite directions during crisis periods.

Similarly, for investors using derivatives products as a hedge on the underlying assets and for risk management, it is advisable that such investors should buy call and put options respectively for assets whose returns move in opposing directions as may be witnessed from the calculated empirical correlation matrix. Furthermore, an accurate quantification of correlations between the returns of various stocks is of practical importance in quantifying the risk of portfolios of stocks, pricing of options and forecasting, V. Plerou et al. [12]. They assert that financial correlation matrices are the key input parameters to Markowitz's [33] classical theory of portfolio optimization problem that aims at providing a recipe for the selection of a portfolio of assets in order that the risk associated with the investment is minimized for a given expected return.

The interesting questions that genuine investors need to answer are how the (implied) volatility, which is a measure of market fluctuations, affects the dynamics of the market or vice versa. It is, therefore, expedient to explore the relationship between the two distinct properties of the market which includes volatility as an index of market fluctuations and the coupling of stocks with one another using the concept of correlation matrix [19]. Correlations amongst the volatility of different assets are very useful, not only for portfolio selection, but also in the pricing of options and certain multivariate econometric models for price forecasting and volatility estimations [34]. They assert that with regards to Black-Scholes [35] option pricing model the variance of portfolio, p of options exposed to Vega risk only is given by

$$Var(p) = \sum_{i,j,k,l} \frac{w_i w_l \Lambda_{ij} \Lambda_{lk} C_{jk}}{v_j v_k \sigma_j \sigma_k} \quad (1)$$

where w_i are the weights in the portfolio, C_{ij} is the correlation matrix for the implied volatility for the underlying assets and the vega matrix Λ_{ij} is defined as

$$\Lambda_{ij} = \frac{\partial p_i}{\partial v_j} \quad (2)$$

with p_i as the price of option i , v_j is the implied volatility of asset underlying option j and σ_i is the standard deviation of the implied volatility v_i . Similarly, Adrian Buss and Grigory Vilkov [36] assert that given a portfolio of N assets, the variance of portfolio σ_{port}^2 can be calculated using the formula

$$\sigma_{port}^2 = \sum_{i=1}^N \sum_{j=1}^N C_{ij} w_i w_j \sigma_i \sigma_j \quad (3)$$

with σ_{port} = annual standard deviation or volatility of the portfolio, σ_i, σ_j = Annual standard deviation or volatility of the asset i and j , w_i, w_j = Weights of asset i and j respectively, and C_{ij} = Correlation coefficient between asset i and j with $C_{ij} = 1$ for $i = j$, $1 \leq i \leq N$. Equation (3) can, therefore, be used in portfolio management. A portfolio that has minimum variance is said to be less risky. In our study, this can be illustrated by assigning some weight to a portfolio consisting of some selected stocks from NSM and then the value of C_{ij} picked from the empirical correlation matrix coefficient of the respective stocks to determine better portfolio choice(s). The variances of the respective assets are as reflected in equation (8) below. Pollet and Wilson [37] proposed a proxy for the calculation of average correlation using only individual stock volatilities and that of the portfolio/index as a whole starting from the definition of a variance related portfolio risk given by:

$$\sigma^2 = \sum_{i=1}^N w_i^2 \sigma_i^2 + 2 \sum_{i=1}^{N-1} \sum_{j>1}^N C_{ij} w_i w_j \sigma_i \sigma_j \quad (4)$$

Black and Scholes propounded an option pricing formula which with the underlying assumptions can be used to calculate the equilibrium price of stock options [38]. One of the assumptions of the model is the constant volatility but from evidence of implied volatility surfaces which have smiles and skews against a flat surface expected predicted by the model, the earlier constant variance assumption throughout the life span of an option is, therefore, untrue. Implied volatility is therefore seen not as a constant but rather a parameter that varies with respect to time to maturity and moneyness or strike price of the option [29]. In the Black-Scholes option pricing model, historical stock price data is used to estimate the volatility parameter which can be plugged into the model to derive the option values. Alternatively, in a bid to overcome the shortcoming witnessed from the constant volatility assumption of Black-Scholes, one may observe the market price of the option, and then invert the option-pricing formula to determine the volatility implied by the option price. This market assessment of the underlying asset's volatility as reflected in the option price is called implied volatility of the option, Stewart Mayhew [39]. Given these developments therefore, the study

of implied volatility and its relation to correlation matrix becomes indispensable in the exploration of methods of risk management and portfolio optimization. Edelman Alan [40] asserts that random matrix theory makes it possible for a comparison between the cross-correlation matrices obtained from a given number of empirical time series data for a period T with an absolutely random matrix W , otherwise known as Wishart matrix of the same size with the empirical correlation matrix in order to obtain some useful information about the market(s) necessary for portfolio optimization and risk management. RMT predictions represent an average over all possible interactions between the constituents of the asset in a given market under consideration. The deviations from universal predictions of RMT obtained from the Wishart matrix is used in identifying the system specific, non-random properties of the system under consideration and such deviations provide information about the underlying interaction of the assets. In other words, we compare the statistics of the cross-correlation coefficients of price fluctuations of stock i and j against a random matrix having the same symmetric properties as the empirical matrix. The RMT is known to distinguish the random and non-random parts of the cross-correlation matrix C and the non-random parts of C which deviates from RMT results is known to provide information regarding the genuine collective behaviour of the stocks under consideration and indeed the entire market at large [13].

The investigation of correlations between price changes of different assets in any given exchange is not only necessary for quantifying the risk in a given portfolio but also of scientific interest to researchers in economics and financial mathematics [41,42]. However, the problem of interpreting the correlations between individual stocks-price changes in a given financial market can be likened to the difficulties experienced by physicists in the fifties, in interpreting the spectra of complex nuclei. Due to the huge amounts of spectroscopic data on the energy levels that were available which were too complex to be interpreted through model calculations, since the nature of the interactions were not known, the concept of Random Matrix Theory (RMT) was developed to take care of the statistics of energy levels of the complex quantum systems [43-45]. Similarly, for financial time series in a stock exchange, the nature of interactions among constituent stock are unknown hence the need to adopt the RMT method in explaining the nature of interactions between the individual stock for use in portfolio optimization and risk management. The estimation of risk and expected returns based on variance and expected returns in a given portfolio that constitute Markowitz's model [46]. In doing this, the period T under consideration has to be relatively large enough when compared with the number of stocks or assets being considered to minimize the noise in the correlation matrix. The two sources of noise envisaged in the use of RMT in investigating the cross-correlations of stocks in a given financial market include the noise from the period length T considered with respect to the number of stock and that resulting from the fact that financial time series of historical return itself is finite or bounded thereby introducing inadvertently estimation errors (noise) in the correlation matrix [1]. Szilard and Kondor [47]

also observe that the effect of noise strongly depends on the ratio $= \frac{N}{T}$, where n is the number

of stocks considered and t the length of the available time series. They note that for the ratio $r = 0.6$ and above, there will be a pronounced effect of noise on the empirical analysis as was

discovered by G. Galluccio et al. [48]; V. Plerou et al. [1]; L. Laloux et al. [31] and that for smaller value of r ($r = 0.2$ or less); the error due to noise drops to an acceptable levels. In

this study, we use the empirical data obtained from NSM, $r =$ the acceptable value of $r. \frac{82}{1018} = 0.08 < 0.2$, thus within

When this is done, if the eigenvalues of the empirical correlation matrix and that of the Wishart matrix lie in the same region without any significant deviations, then the stocks are said to be uncorrelated and therefore no information or deduction can be made about the nature of the market since it is the deviations of the eigenvalues of the correlation matrix from that of the Wishart matrix that carries information about the entire market. However, if on the contrary there exists at least one eigenvalue lying outside the theoretical bound of the eigenvalues in the empirical correlation matrix obtained from the stock market returns, then the deviating eigenvalue(s) is(are) known to carry information about the market under consideration.

2. Data

The Data set is comprised of the daily closing prices of 82 stocks listed in the Nigerian Stock Market, NSM from 3rd August 2009 to 26th August daily closing returns after removing assets that were delisted, that did not trade at all or are partially traded in the period under review. The assets considered are drawn from the Agriculture, Oil and Gas, Real Estates/Construction, Consumer Goods and Services, Health care, ICT, Financial Services, Conglomerates, Industrial Goods, and Natural Resources.

For the values of the daily asset prices to be continuous and to minimize the effect of thin trading, we will remove the public holidays in the period under consideration and to reduce noise in the analysis, market data for the present day is assumed to be the same with the previous day for cases where there are no information on trade for any particular asset on a given date. Also, we eliminate stocks that infrequently traded within the period under review. Let $\delta' f \delta' - (\delta' i)$ be the closing price on a given day t , for stock i and define the natural logarithmic return of the index as

$$r_i(t) = \ln \frac{P_i(t+1)}{P_i(t)} \quad (5)$$

where $r_i(t)$ is the number of observations in the two exchanges, NSM and JSE.

3 Theoretical Backgrounds

3.1 Computing Volatility

We calculate the price changes of assets in the two markets over a time scale Δt which is equivalent to one day and denote the price of i at a time t as $s_i(t)$ with the corresponding price change or logarithmic returns $G_i(t)$ over time scale Δt as

$$G_i(t) = \ln [S_i(t + \Delta t)] - \ln [S_i(t)] \quad (6)$$

We quantify the volatility in the respective asset return as a local average of the absolute value of daily returns of indices in an appropriate time window of T days as

$$v = \frac{\sum_{t=1}^{T-1} |G_i(t)|}{T-1} \quad (7)$$

To standardize the values of obtained from equation (6) above for all values of i , we normalize $G(i)_t$ as follows:

$$g(i)_t = \frac{G(i)_t - \langle G(i)_t \rangle}{\sigma_i} \quad (8)$$

where $\sigma_i = \sqrt{\langle G(i)_t^2 \rangle - \langle G(i)_t \rangle^2}$ and $\langle \dots \rangle$ represents the average in the period studied.

From real time series return data, we can calculate the element of $N \times N$ correlation matrix C as follows

$$C_{ij} = \langle g(i)g(j) \rangle = \frac{\langle [G_i(t) - \langle G_i \rangle][G_j(t) - \langle G_j \rangle] \rangle}{\sqrt{[\langle G_i^2 \rangle - \langle G_i \rangle^2][\langle G_j^2 \rangle - \langle G_j \rangle^2]}} \quad (9)$$

C_{ij} lies in the range of the closed interval $-1 \leq C_{ij} \leq 1$, with $C_{ij} = 0$ means there is no correlation, $C_{ij} = -1$ implies anti-correlation and $C_{ij} = 1$ means perfect correlation for the empirical correlation matrix gotten from the implied volatility surface.

3.2 Eigenvalue spectrum of the correlation matrix

As stated earlier, our aim is to extract information about the cross-correlation from the empirical correlation matrix C . To this end we are going to compare the properties of C with those of a random matrix [T. Colon et al. [49]; L. Laloux et al. [31]; V. Plerou et al. [6]; P. Gopikrishnan [32]; V. Plerou et al. [14]. It can be shown from Sharifi, S [8] that the empirical correlation matrix C can be expressed as

$$C = \frac{1}{L} G G^T \quad (10)$$

where G is the normalized $N \times L$ matrix and G^T is the transpose of G . This empirical correlation will be compared with a random Wishart matrix (random matrix) R given by:

$$R = \frac{1}{L} A A^T \quad (11)$$

to classify the information and noise in the system, T. Colon et al. [49] and P. Gopikrishnan et al. [32], where A is an $N \times L$ matrix whose entries are independent identically distributed random variables that are normally distributed and have zero mean and unit variance. In our bid to use the random matrix theory in portfolio optimization and (derivative) assets risk management, we should be conversant with the universal properties of random matrices. Wilcox et al. [28] assert that there are four underlying properties of random matrices which include (a) Wishart distribution eigenvalues from the correlation matrix, (b) Wigner surmise for eigenvalue spacing (c) the distribution of eigenvector components of the corresponding eigenvalues and finally (d) Inverse participation ratio for eigenvector components of the resulting correlation matrix.

[50-52], and A. Edelman [40] assert that the statistical properties of R are known and that in particular for the limit as $N \rightarrow \infty$, and $L \rightarrow \infty$ we have that $Q = \frac{L}{N} (\geq 1)$ is fixed and that

the probability function $P_{rm}(\lambda)$ of eigenvalues λ of the random correlation matrix R is given by

$$P(\lambda) = \frac{Q}{2\pi\sigma^2} \frac{\sqrt{(\lambda_{max}-\lambda)(\lambda-\lambda_{min})}}{\lambda} \quad (12)$$

for λ such that $\lambda_{min} \leq \lambda \leq \lambda_{max}$, where σ^2 is the variance of the elements of A . Here $\sigma^2 = 1$ and λ_{min} and λ_{max} satisfy

$$\lambda_{max/min} = \sigma^2 (1 + \frac{1}{Q} \mp 2\sqrt{1/Q}) \quad (13)$$

The values of λ from equation (11) that satisfy (12) and (13) are called the Wishart distribution of eigenvalues from the correlation matrix. These values of λ as stated before determining the bounds of theoretical eigenvalue distribution. When the eigenvalues of empirical correlation matrix C are beyond these bounds, they are said to deviate from the random matrix bounds and are therefore supposed to carry some useful information about the market, [25].

The distribution of eigenvalue (surmise of eigenvalue) spacing was introduced as the required test for the case when there are not significant deviations of the empirical eigenvalue distribution to that of the random matrix prediction, [28]. When the eigenvalues so obtained from the correlation matrix do not deviate significantly from the predictions of the RMT we apply the so-called Wigner surmise for eigenvalue spacing otherwise called Gaussian orthogonal ensemble Plerou et al. [14] is given by

$$P(s) = \frac{s}{2\pi} \exp\left(-\frac{s\pi^2}{4}\right), \quad (14)$$

where $(\lambda_{i+1} - \lambda_i)/d$ and d denotes the average of the differences $\lambda_{i+1} - \lambda_i$ as i varies.

3.3 Distribution of eigenvector component

The concept that low lying eigenvalues are really random can also be verified by studying the statistical structure of the corresponding eigenvectors. The j th component of the eigenvector corresponding to each eigenvalue λ_α will be denoted by $v_{\alpha,j}$ and then normalized such that $\sum_{j=1}^N v_{\alpha,j}^2 = N$. Plerou et al. [6] assert that if there is no information contained in the eigenvector $v_{\alpha,j}$, one expects that for a fixed α , the distribution of $u = v_{\alpha,j}$ (as j is varied) is a maximum entropy distribution. This, therefore, leads to what is called Porter-Thomas distribution in the theory of random matrices written as

$$p(u) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{u^2}{2}\right) \quad (15)$$

It has been found that the eigenvector components $v_{\alpha,j}$ for $\alpha=1, 2, 3 \dots n$ of an eigenvector

v_α are normally distributed with zero mean and unit variance, [53]. The distribution so obtained from (14) above are expected to fit well the histogram of the eigenvector except for

those corresponding to the highest eigenvalues which lie beyond the theoretical value of λ_{max} , [6].

3.4 Inverse participation ratio

Guhr, T. et al. [53] assert that to quantify the number of components that participates significantly in each eigenvector, we use inverse participation ratio. Inverse participation ratio (IPR) shows the degree of deviation of the distribution of eigenvectors from RMT results and distinguishes one eigenvector with approximately equal components with another that has a small number of large components. For each eigenvector v_α , V. Plerou et al. [14] defined the inverse participation ratio as

$$I_\alpha = \sum_{j=1}^N (v_\alpha^j)^4 \quad (16)$$

where N is the number of the time series (the number of implied volatility considered) and

hence the number of eigenvalue components and v_α^j is the j th component of the eigenvector v_α . There are two limiting cases of I_α (i) If an eigenvector v_α has an identical component,

$$v_\alpha(j) = \frac{1}{\sqrt{N}}, \text{ then } I_\alpha = \frac{1}{N}$$

and (ii) For the case when v_α has one element with $v_\alpha(j) = 1$ and the remaining components zero, then $I_\alpha = 1$.

Therefore, the IPR can be illustrated as the inverse of the number of elements of an eigenvector that are different from zero that contribute significantly to the value of the eigenvector. A. Utsugi et al. [54] in their study of the RMT assert that the expectation of the IPR is given by

$$\langle I_\alpha \rangle = N \int_{-\infty}^{\infty} [v_\alpha(j)]^4 \frac{1}{\sqrt{2\pi N}} \exp\left(-\frac{[v_\alpha(j)]^2}{2N}\right) dv_\alpha(j) = \frac{3}{N} \quad (17)$$

since the kurtosis for the distribution of eigenvector components is 3.

3.5 Algorithm for Calculating Realistic Implied Correlation Matrix, R^Q

Kawee and Nattachai Numpacharoen [55] defined a valid empirical correlation matrix from an identity $n \times n$ matrix as a matrix with the following properties: (a) All the diagonal entries must be one which is the case for the empirical correlation matrix obtained from the sample of stocks considered with the NSM in this paper (b) Non-diagonal entries of C_{ij} are real numbers in the closed interval $-1 \leq C_{ij} \leq 1$ (c) The empirical correlation matrix is symmetric (d) The empirical correlation matrix must be positive (semi) definite to accommodate matrix decomposition for some desired purposes like Monte-Carlo simulation Kawee Numpacharoen [2]. They further stated that when the empirical correlation matrix are not identical as is the case with the matrix derived from the asset return distribution of stocks

selected from NSM, the implied volatility of the portfolio σ_{port}^Q is given by

$$(\sigma_{port}^Q)^2 = W * S^Q * C^Q * S^Q * W' \quad (18)$$

Similarly, if σ_{port}

is the implied volatility of the portfolio obtained from C^P then it can be described as

$$(\sigma_{port}^P)^2 = W * S^Q * C^P * S^Q * W' \quad (19)$$

where $W = [w_1 \dots w_n]$ are the weights of the respective stocks in the portfolio;

$$S^Q = \begin{bmatrix} \sigma_1^Q & 0 \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \sigma_n^Q \end{bmatrix}$$

is a diagonal matrix got from the implied standard deviation of the respective assets being considered.

$$C^Q = \begin{bmatrix} 1 & C_{2,1}^Q \dots C_{n-1,1}^Q & C_{n,1}^Q \\ \vdots & \ddots & \vdots \\ C_{n,1}^Q & C_{n,2}^Q \dots C_{n-1,n}^Q & 1 \end{bmatrix} \text{ is the desired realistic implied correlation matrix.}$$

$$\text{or} \quad (\sigma_{port}^Q)^2 = \sum_{i=1}^N \sum_{j=1}^N C_{ij}^Q w_i w_j \sigma_i^Q \sigma_j^Q \quad (18a)$$

$$\rightarrow (\sigma_{port}^Q)^2 = \sum_{i=1}^N w_i^2 (\sigma_i^Q)^2 + 2 \sum_{i=1}^{N-1} \sum_{j>i}^N C_{ij}^Q w_i w_j \sigma_i^Q \sigma_j^Q \quad (18b)$$

Buss and Vilkov [36] assert that to identify $N \times (N - 1)/2$ correlations that satisfy equation (18a), we propose the following parametric form for implied correlations:

$$C^Q = C^P - \varphi * (I_{n \times n} - C^P) \quad (20)$$

where, C^Q is the expected correlation under the objective measure and φ is the parameter to be identified.

By substituting equation (20) into equation (18) we shall have:

$$\varphi = - \frac{(\sigma_{port}^Q)^2 - W * S^Q * C^P * S^Q * W'}{W S^Q (I_{n \times n} - C^P) S^Q W'} \quad (21)$$

and from equations (18) and (19) is equivalent to:

$$\varphi = \frac{(\sigma_{port}^Q)^2 - \sum_{i=1}^N \sum_{j=1}^N C_{ij}^P w_i w_j \sigma_i^Q \sigma_j^Q}{\sum_{i=1}^N \sum_{j=1}^N w_i w_j \sigma_i^Q \sigma_j^Q (I_{n \times n} - C^P)} \quad (22)$$

Buss and Vilkov [36] impose a restriction on the values φ to be in the region $-1 < \varphi \leq 0$ for it to satisfy the technical conditions on the correlation matrix which includes that all the correlation $C_{i,j}$, do not exceed one and that the correlation matrix is positive definite.

Now we consider the cases when $\varphi > 0$ and this occurs when $\sigma_{port}^P > \sigma_{port}^Q$ since $I_{n \times n} -$

$C^P \geq 0$ and this could make the realistic correlation matrix C^Q to be invalid when $\sigma_{port}^P <$

[55]. He proposes a formula for valid correlation matrix that will take care of this σ_{port}^Q shortcoming as stated below. Given any two valid correlation matrices C and D of

dimensions $n \times n$ then, there exists another valid correlation matrix F of the same dimension such that

$$F = w * C + (1 - w) * D \quad (23)$$

where w is a real number in the closed interval $0 \leq w \leq 1$.

It therefore depends on the nature of the inequality existing between σ_{port}^P and σ_{port}^Q respectively that will inform our decision on the equivalent upper or lower bound equicorrelation matrix C to be used in obtaining a realistic implied correlation matrix. The corresponding equicorrelation matrices are represented by $I_{n \times n}$ for upper equicorrelation

matrix and $L_{n \times n}$ matrix whose entries are $-\frac{1}{n-1}$ for $i \neq j$ and 1 for $i = j$ as the lower equicorrelation matrix.

Replacing equation F by C^Q and D by C^P in (23) we will obtain

$$C^Q = C^P + w * (C - C^P) \quad (23a)$$

and from equations (18) and (19) we shall have:

$$w = \frac{(\sigma_{port}^Q)^2 - (\sigma_{port}^P)^2}{W S^Q (C - C^P) S^Q W'} \quad (24)$$

This approach is useful in assigning the respective weights to different assets in our portfolio as seen in Table 1 which will help in maximizing the returns and minimizing the risk on our portfolio of investments. To this effect and as an empirical demonstration, we therefore use the correlation matrix obtained from NSM stock returns on the various assets considered in the NSM from 2009 to 2013 for some selected assets. The assets are 7UP, ABCTransport, Access Bank, AgLevent, AIICO Insurance and Air service. We, therefore, want to compute the realistic empirical correlation matrix for some assets already considered in the RMT as below:

	7UP	ABCTrans	Access	AGLevent	AIICO	AIRService
7UP	1	-0.051	-0.003	0.003	-0.001	0.017
ABCTrans	-0.051	1	0.057	0.107	0.026	-0.001
Access	-0.003	0.057	1	0.042	0.166	0.005
AGLevent	0.003	0.107	0.042	1	0.027	0.001
AIICO	-0.001	0.026	0.166	0.027	1	-0.007
AIRService	0.017	-0.001	0.005	0.001	-0.007	1

Table1: Empirical correlation matrix from NSM price return

4.0 Empirical Result and data analysis

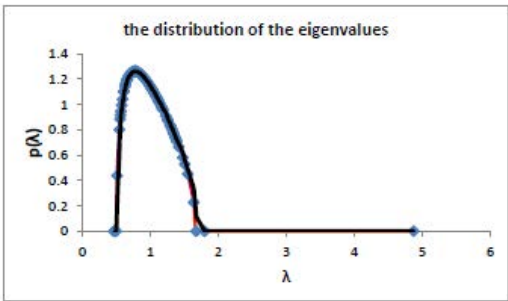
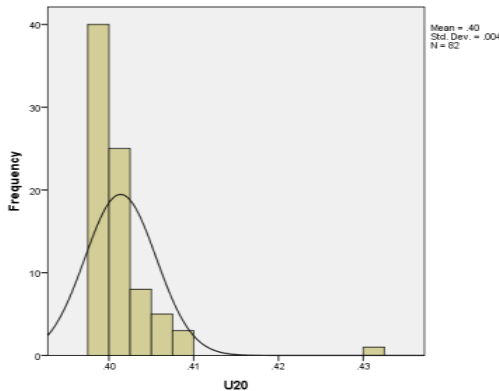
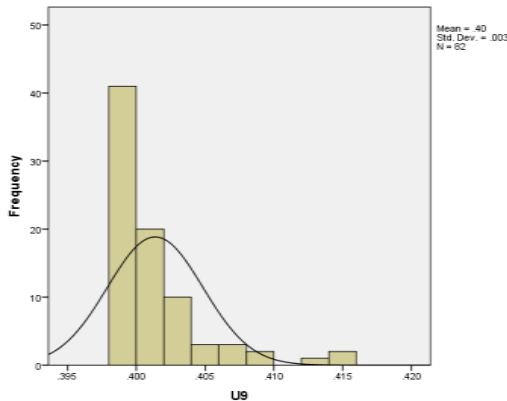
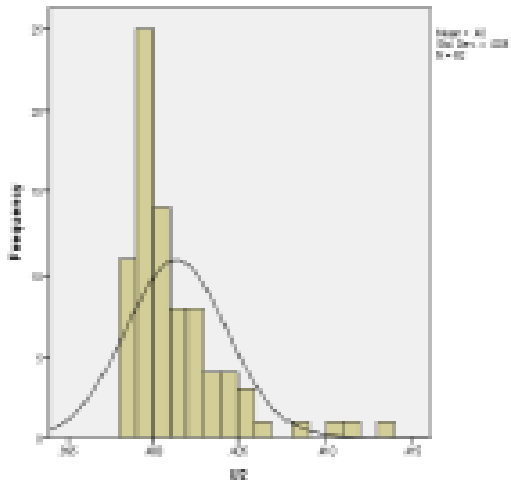
4.1 Eigenvalue analysis

We took a sample study of eighty-two ($N = 82$) stocks from the Nigerian stock exchange totaling $L = 1019$ daily closing prices and that of the Johannesburg stock exchange, JSE we had a sample study of thirty-five ($N' = 35$) stocks with a total of $L' = 1148$. For NSM the theoretical eigenvalue bounds are respectively $\lambda_- = 0.51$ and $\lambda_+ = 1.65$ as minimum and

maximum values with $Q = \frac{L}{N} = 12.41$. Further from the calculation the market value shows

that the largest eigenvalue $\lambda_1 = 4.87$ which is approximately three times larger than the predicted RMT of value (1.64).

Figure 1: Theoretical (Marcenko-Pastur) empirical eigenvalues for NSM.



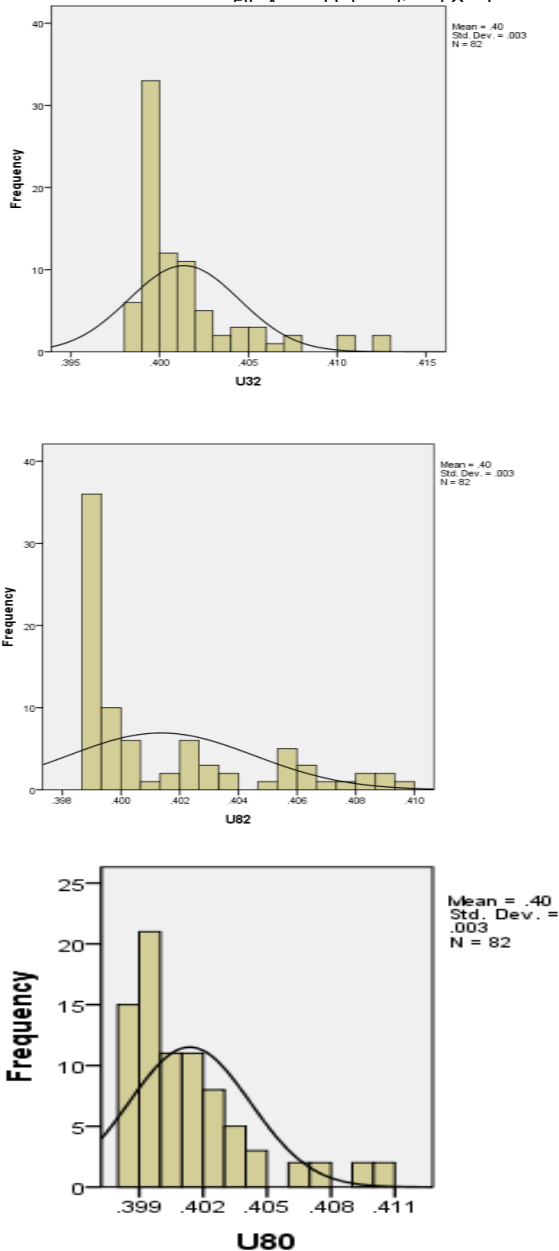


Figure 2: Distribution of eigenvector components of stocks in NSM

Figure 2 above represent the distribution of eigenvectors for the various eigenvalues in the empirical correlation matrix. The diagram labeled U82 represents an eigenvector component for deviating eigenvalue in the theoretical region where as the other 4 are the eigenvector components of the eigenvalue within the regions predicted from the random matrix theory.

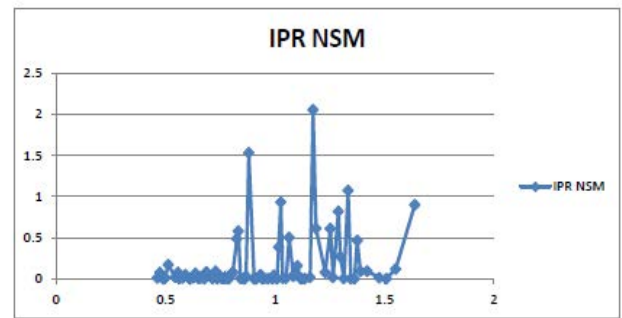


Figure 3: Inverse participation ratio and their ranks for NSM.

The inverse participation ratio (IPR) is the multiplicative inverse of the number of eigenvector components that contribute significantly to the eigenmode, [14]. For the largest eigenvalue (deviating from the RMT bounds) almost all the stocks contribute to the corresponding eigenvector thereby justifying treating this eigenvector as the market factor. The eigenvector corresponding to other deviating eigenvalues also exhibits that their corresponding stocks contribute slightly to the overall market features in the NSM. The

average IPR value is around $\frac{3}{82}$ for NSM larger than would be expected $\frac{1}{N} = 0.01$ for NSM, if all components contributed to each eigenvector, [49]. The remaining eigenvectors appear to be random with some deviations from the predicted value of $\frac{3}{N} = 0.04$ possibly as a result of the existence of fat tails and high kurtosis of the return distributions.

4.2 Realistic Implied Correlation matrix computations:

Suppose we had the following weights and implied volatility (computed from option prices) for the under listed assets drawn from the Nigerian Stocks Market.

Asset	weight	implied volatility	Corr. Coeff.					
7UP	15%	23%	1.0000	-0.0508	-0.0026	0.0032	-0.0014	0.0157
ABCTrans	28%	26%	-0.0508	1.0000	0.0565	0.1073	0.0264	-0.0006
Access	10%	30%	-0.0026	0.0565	1.0000	0.0418	0.1658	0.0052
AgLevent	35%	15%	-0.0032	0.1073	0.0418	1.0000	0.0267	0.0012
AIICO	23%	34%	-0.0014	0.0264	0.1656	0.0267	1.0000	-0.0073
AirService	18%	36%	0.0157	-0.0006	0.0052	0.0012	-0.0073	1.0000

Table 2: Empirical correlation matrix of some assets considered in NSM with their assumed weights and Implied volatilities.

$$C^P = \begin{bmatrix} 1.0000 & -0.0508 & -0.0026 & 0.0032 & -0.0014 & 0.0157 \\ -0.0508 & 1.0000 & 0.0565 & 0.1073 & 0.0264 & -0.0006 \\ -0.0026 & 0.0565 & 1.0000 & 0.0418 & 0.1656 & 0.0052 \\ 0.0032 & 0.1073 & 0.0418 & 1.0000 & 0.0267 & 0.0012 \\ -0.0014 & 0.0264 & 0.1656 & 0.0267 & 1.0000 & -0.0073 \\ 0.0157 & -0.0006 & 0.0052 & 0.0012 & -0.0073 & 1.0000 \end{bmatrix}$$

The eigenvalues of C^P are $[1.2205, 0.8308, 0.8808, 1.0675, 0.9843, 1.0148]'$.

Thus the minimum eigenvalue of $C^P = 0.8308$ which shows that C^P is a valid correlation matrix. Therefore, to estimate the realistic implied correlation matrix C^Q from the assumed implied volatility for the given a portfolio consisting of six assets, we assume that the implied volatility of portfolio $\sigma_{port}^Q = 0.28$.

From Table 1: The weights of the respective assets $W = [0.15 \ 0.28 \ 0.10 \ 0.35 \ 0.23 \ 0.18]'$;

$$S^Q = \begin{bmatrix} 0.23 & 0.00 & 0.00 & 0.00 & 0.00 & 0.00 \\ 0.00 & 0.26 & 0.00 & 0.00 & 0.00 & 0.00 \\ 0.00 & 0.00 & 0.30 & 0.00 & 0.00 & 0.00 \\ 0.00 & 0.00 & 0.00 & 0.15 & 0.00 & 0.00 \\ 0.00 & 0.00 & 0.00 & 0.00 & 0.34 & 0.00 \\ 0.00 & 0.00 & 0.00 & 0.00 & 0.00 & 0.36 \end{bmatrix}$$

We now use equation (19) and the respective values of W, S^Q, C^P , to calculate σ_{port}^P .

$$\sigma_{port}^P = \text{SQRT}(W * S^Q * C^P * S^Q * W') = 0.1546.$$

Since $0.28 > 0.15 \rightarrow \sigma_{port}^Q >$

σ_{port}^P , therefore we shall replace C in equation (24) by an equivalent identity 6×6 equicorrelation matrix to obtain the value of w :

$$w = \frac{(\sigma_{port}^Q)^2 - (\sigma_{port}^P)^2}{W * S^Q * (I_{6 \times 6} - C^P) * S^Q * W'} = \frac{0.0545}{W * S^Q * (I_{6 \times 6} - C^P) * S^Q * W'} = \frac{0.0545}{0.0914} = 0.5963$$

Therefore, $C^Q = C^P + 0.5963 * (I_{6 \times 6} - C^P)$

$$= \begin{bmatrix} 1.000 & 0.576 & 0.595 & 0.598 & 0.596 & 0.603 \\ 0.576 & 1.000 & 0.619 & 0.640 & 0.607 & 0.596 \\ 0.595 & 0.619 & 1.000 & 0.613 & 0.663 & 0.598 \\ 0.598 & 0.640 & 0.613 & 1.000 & 0.607 & 0.597 \\ 0.596 & 0.607 & 0.663 & 0.607 & 1.000 & 0.593 \\ 0.603 & 0.596 & 0.598 & 0.597 & 0.593 & 1.000 \end{bmatrix}$$

The eigenvalues of $C^Q = [4.034, 0.445, 0.430, 0.398, 0.357, 0.336]'$ from where we obtain the minimum eigenvalue to be 0.3359

showing that C^Q is also positive semi-definite.

We now verify our solution from equation (18) to compute the variance of portfolio using the obtained realistic implied correlation matrix C^Q gotten above:

$$\sigma_{port}^P = \text{SQRT}(W * S^Q * C^Q * S^Q * W') = 0.28 \text{ as required.}$$

Contribution to knowledge:

The implied correlation matrix is applicable in hedging risks associated with foreign exchange. Large corporations are always very interested in hedging their currency exposures by using a basket of options instead of taking separate put options for the respective countries where they have their investments [56]. This will help guard against the unnecessary losses they might incur in the event of rising value of the domestic currency where they have their investments. Companies that are therefore exposed to a variety of currency fluctuations find it profitable to directly hedge their aggregate risk by using a basket of options made possible through the use of estimated implied correlation matrix in the form of basket of options [57]. For a manufacturing firm in the United States that sources its raw materials in Nigeria, Ghana, and South Africa and pays for its operation in those countries in local currencies will be exposed to exchange rate risk. To hedge against the risk of falling United States dollars against Naira, Cedi, and Rand, the manufacturing firm has to use a basket of option in its risk management strategy. The company could therefore directly buy an option on a basket of currencies at a lower price than it can purchase through separate options on the individual currencies. This is possible through the use of historical return time series correlation, as we have done with some stocks in the NSM, and the major concern will then be the amount of weight to be assigned to the individual stocks (or currencies). For optimal portfolio on the investment, we need to predict correctly the future correlation of the respective option values by observing the correlation throughout the life span of the option.

5. Conclusion and hints on future work

The analysis of the correlation and structure of stock market returns has provided the necessary data for a hypothetical analysis of implied correlation in the NSM, carried out in this research using the concept of RMT. This foregrounds research on comparative derivative pricing in the NSM, especially on foreign exchange futures. Marcenko-Pastur eigenvalue distribution predicted that the theoretical eigenvalues should be in the range of 0.52 and 1.65 for NSM. It was observed that 6 out of 82 stocks considered that have their corresponding eigenvalues lie outside this theoretical bound of eigenvalues, therefore, 89% of the information from the return distributions is purely random thereby leaving us with the alternative hypothesis of the RMT which states that the information on the market lies on the deviating eigenvalues which imply then that for NSM the actual market characteristic lies with only 11% of the stocks considered.

As stated earlier, these correlation matrices contain some relevant information for options pricing and hedging, [58]. The realistic implied correlation matrix $\hat{\rho}$, has positive coefficients meaning that the respective stock move in the same direction hence the diversification method in the portfolio is not an optimal portfolio strategy. It is, therefore, better to invest in some derivative products like call and or put option to hedge against the risk on the portfolio for the hypothetical weight and implied volatility used in the estimated implied correlation matrix.

We also noted that the concept of implied correlation could be used in options trading and hedging the risks associated with the portfolio of investment including the use of a basket of options in hedging foreign exchange risk. In this regard, it is therefore open to further work for the investigation of the use of the basket of an option to hedge against exchange rate risk especially in emerging markets like Nigeria and South Africa. As Nigeria has already started trading on foreign exchange futures, it is pertinent therefore to note that soon market data will be available for the empirical application of the concept of implied correlation matrix for the hedging of exchange risk in Nigeria.

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* Corresponding author:

Emails: thomas.c.urama@student.shu.ac.uk or
chinweurama@yahoo.com

Statistics, Information Modelling & Financial Mathematics Research Group, Department of Engineering & Mathematics, Sheffield Hallam University, S1 1WB, United Kingdom. The researchers are grateful to Tertiary Education Trust Fund Nigeria for funding this research through a sponsorship of their doctoral programme at Sheffield Hallam University, UK. We are also grateful to Staff and Management of Sheffield Hallam University, the UK for providing the facilities and very conducive environment for research and learning.

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